Planet of The Aves: An Immersive Educational Application for Learning Bird Habitats and Lifecycle

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***Abstract*—The advancement of technology in education has introduced new ways to create more interactive learning experiences, especially in biology education. Nevertheless, learning materials about bird habitats, adaptation, and reproduction lifecycle are still often delivered using text and static images, which can limit user engagement and understanding. Therefore, this project proposes an immersive educational application called "Planet of The Aves" to provide a more interactive learning environment through habitat exploration, bird information, interactive 3D visualization, lifecycle interaction, educational popup features, and quizzes.**

The application is developed using the Agile Software Development Life Cycle (SDLC) approach to support gradual and flexible development processes. Several immersive elements such as environmental visualization, interactive objects, ambience audio, and 3D bird exploration are integrated into the application to enhance user interaction and educational experiences. Through these features, the application is expected to assist users in understanding bird habitats, adaptation processes, and reproduction lifecycle in a more engaging and interactive manner.

Keywords—Immersive Learning, Educational Application, Aves, Interactive Learning, Mobile Application, Agile SDLC, 3D Exploration

I. INTRODUCTION

A. Background

The development of technology in education has significantly changed the way students learn and interact with learning materials [1]. Traditional learning methods that rely heavily on textbooks and static visual materials are often considered less engaging, especially in subjects that require visualization and exploration, such as biology [2]. One of the topics in biology that contains many interesting concepts is aves or birds, particularly their habitats, adaptations, and reproduction lifecycle.

Birds have various habitat characteristics and survival adaptations depending on their environment, such as tropical forests, wetlands, savannahs, and arctic areas. However, learning about aves is still commonly delivered through text-based explanations and two-dimensional images, making it difficult for users to fully understand the environmental conditions and lifecycle processes of different bird species. As a result, users may lose interest and have limited interaction with the learning material [3].

To solve this problem, immersive learning technology can be used to create a more interactive and engaging educational experience [4]. Immersive learning combines multimedia, interaction, and virtual exploration to improve user understanding and participation during the learning process [5]. By integrating interactive elements such as 3D models, environmental visualization, educational popups, and quizzes, users can experience a more realistic and enjoyable learning activity [6].

Based on these conditions, this project proposes an immersive educational application called "Planet of The Aves." The application is designed to help users learn about bird habitats, species adaptation, and reproduction lifecycle through interactive exploration and immersive learning experiences. Users can select habitats, explore bird information, interact with 3D bird objects, and learn the lifecycle of aves in a more visual and engaging way.

B. Problem Statement

Birds have various habitat characteristics and survival adaptations depending on their environment, such as tropical forests, wetlands, savannahs, and arctic areas. However, learning about aves is still commonly delivered through text-based explanations and two-dimensional images, making it difficult for users to fully understand the environmental conditions and lifecycle processes of different bird species. As a result, users may lose interest and have limited interaction with the learning material.

Based on the background above, several problems can be identified:

- Learning media about aves is still mostly text-based and less interactive.
- Users often find it difficult to visualize bird habitats, adaptations, and reproduction processes through conventional learning methods.

- Educational applications with immersive interaction and 3D exploration for aves learning are still limited.Traditional biology learning methods may reduce user interest and engagement during the learning process.

C. Proposed Solutions

To address these problems, this project proposes the development of an immersive educational application called "Planet of The Aves." The application provides interactive learning features that allow users to explore different bird habitats and study bird reproduction lifecycles through immersive interaction.

The application includes several features such as habitat selection, bird information pages, interactive 3D bird exploration, lifecycle interaction, educational popups, and quizzes. By combining immersive environments, 3D interaction, and educational content, the application is expected to improve user engagement and understanding of aves-related materials.

D. Objective

The application includes several features such as habitat selection, bird information pages, interactive 3D bird exploration, lifecycle interaction, educational popups, and quizzes. By combining immersive environments, 3D interaction, and educational content, the application is expected to improve user engagement and understanding of aves-related materials.

The objectives of this project are:

- To develop an immersive educational application for learning aves habitats and reproduction lifecycle.
- To provide interactive learning experiences using 3D exploration and immersive environments.
- To increase user interest and engagement in biology learning materials.
- To help users understand bird adaptation, habitat characteristics, and lifecycle processes more effectively.

E. Scope of Project

This project focuses on the development of an immersive educational application about aves using selected bird species from four different habitats, Which Tropical Forest, Wetland, Savannah and Arctic Area. The application includes; habitat exploration, bird information, interactive 3D exploration, lifecycle learning, educational popup, quiz features

The application is designed for educational purposes and targeted for students and general users interested in learning about birds and biology.

II. Literature Review

A. Immersive Learning

Immersive learning is a learning approach that utilizes interactive technology and multimedia elements to create engaging educational experiences [7]. This method allows users to actively participate in the learning process through visualization, interaction, and virtual exploration. Compared to conventional learning methods, immersive learning can improve motivation, understanding, and user engagement during educational activities [8].

Immersive learning environments are commonly implemented using technologies such as virtual reality, augmented reality, 3D visualization, and interactive multimedia systems. These technologies help users understand educational content more effectively by providing realistic and interactive experiences [9].

In this project, immersive learning is implemented through interactive habitat exploration, 3D bird visualization, educational popups, and lifecycle interaction features to improve user understanding of aves habitats and reproduction processes.

B. Educational Application

Educational applications are digital systems designed to support learning activities using interactive technology [10]. Educational applications can provide flexible and engaging learning experiences through multimedia content such as images, audio, animation, quizzes, and interactive objects.

Modern educational applications focus on user-centered learning approaches, where users can actively interact with educational materials rather than only reading static information. Interactive educational systems are considered more effective in increasing user participation and improving learning outcomes [11].

Planet of The Aves is categorized as an educational application because it provides educational content related to bird habitats, adaptation, and reproduction lifecycle through immersive and interactive learning features.

C. Interactive Multimedia Learning

Interactive multimedia learning combines various multimedia components such as text, images, audio, video, animation, and interaction in one learning system [8]. The purpose of multimedia learning is to improve educational delivery and make learning materials easier to understand.

Interactive multimedia allows users to explore learning materials actively through interaction and visualization. Features such as clickable objects, quizzes, audio feedback, and visual exploration can increase user engagement and understanding [12].

In this project, interactive multimedia concepts are implemented through habitat visualization, interactive educational popups, immersive 3D exploration, and quiz features.

D. Aves Habitat and Adaptation

Aves or birds are vertebrate animals that can adapt to various environmental conditions depending on their habitats. Different habitats require different survival adaptations related to climate, food availability, and environmental characteristics.

Bird species living in tropical forests, wetlands, savannahs, and arctic regions have unique physical and behavioral adaptations that help them survive in their environments. For example, birds in tropical forests often have strong flying abilities and colorful feathers, while birds in arctic areas have thick feathers to survive cold temperatures.

Understanding habitat characteristics and adaptation processes is important in biology education because it explains how organisms interact with their environments. This project uses selected bird species from several habitats as educational materials to help users understand aves adaptation and reproduction more effectively.

E. SDLC Agile Method

Software Development Life Cycle (SDLC) is a systematic process used in software development. One of the commonly used SDLC methods is Agile [13]. Agile is a flexible software development methodology that focuses on iterative development, collaboration, and continuous improvement.

The Agile method allows development teams to adapt to changes during the development process. This method is suitable for projects that may require feature updates, design improvements, or additional requirements throughout development stages.

In this project, Agile is used because the development of Planet of The Aves involves immersive interaction, UI/UX improvement, and continuous feature adjustments during application development.

F. Interactive 3D Exploration

Interactive 3D exploration is a technology that enables users to interact with three-dimensional virtual objects through zooming, rotating, and object exploration [14]. Compared to static images, 3D interaction provides more realistic visualization and immersive learning experiences.

Interactive 3D systems are widely used in educational applications because they can improve user understanding of complex concepts through direct interaction and exploration [9]. In immersive learning environments, 3D visualization can increase educational engagement and make learning activities more interesting.

In this project, interactive 3D exploration is implemented to allow users to explore bird models, habitats, and lifecycle objects interactively within the application.

III. METHODS

A. SDLC Method

This project uses the Agile Software Development Life Cycle (SDLC) method during the application development process. Agile is chosen because it allows the development process to be carried out incrementally and provides flexibility when adjustments or improvements are needed during implementation [15]. The method also supports collaboration among team members and enables continuous evaluation throughout the development stages.

The use of Agile is considered appropriate for this project because Planet of The Aves includes immersive interaction, interface design, and educational features that may continue to evolve during development. Through this method, the application can be improved gradually based on development progress and testing results.

B. Planning

The planning stage focuses on determining the main concept of the application, project objectives, target users, and application features. Planet of The Aves is designed as an immersive educational application that introduces bird habitats, adaptation, and reproduction lifecycle through interactive learning experiences.

The application targets students and general users who are interested in biology learning. Four habitat categories are implemented in the application, namely Tropical Forest, Wetland, Savannah, and Arctic Area. Several bird species are selected from each habitat to represent different environmental conditions and survival adaptations.

At this stage, the development team also determines the application flow and divides tasks among team members according to their responsibilities.

C. Requirement Analysis

Requirement analysis is conducted to identify the system requirements before the implementation process begins. The requirements are divided into functional requirements and non-functional requirements.

The functional requirements include features that allow users to select habitats, choose bird species, access educational information, explore lifecycle processes, interact with 3D bird objects, and complete quizzes provided by the application.

Meanwhile, the non-functional requirements focus on system performance and usability. The application is expected to provide responsive navigation, immersive interaction, stable performance, and user-friendly interfaces to support interactive learning activities.

D. System Design

The design stage focuses on creating the application structure, navigation flow, and interface design. The application flow is designed to provide a simple and interactive learning experience for users.

The application flow begins from the Homepage or Choose Environment page, followed by the Environment Information Page, Bird Selection, Bird Information Page, Interactive 3D Exploration, Interactive Lifecycle Exploration, Quiz Section, and Explore Again or End Session.

The UI/UX design is created using Figma. Several interface components such as layouts, buttons, popup information, habitat cards, and quiz pages are designed during this stage. In addition, immersive environments for each habitat are also prepared by adjusting visual elements, colors, and ambience sounds to improve user interaction and educational experiences.

E. Asset Preparation

Several supporting assets are prepared to support immersive learning features within the application. The assets include 3D bird models, nests, eggs, baby birds, habitat backgrounds, icons, buttons, and ambience audio.

These assets are used to create interactive visualization and immersive exploration features. Environmental audio and visual elements are also implemented to make the educational experience more engaging and realistic.

F. Application Development

The development stage focuses on implementing the planned features into the application system. Several pages and features are developed, including the homepage, habitat selection page, bird information page, interactive 3D exploration, educational popup system, lifecycle interaction, and quiz section.

The application is developed using Flutter or React Native with supporting technologies such as Dart, GitHub, Figma, and Three.js or model viewer libraries. GitHub is used as a version control platform and collaboration tool during the development process.

G. Testing and Improvement

Testing is performed to ensure that all application features and navigation systems function properly. The testing process includes navigation testing, popup interaction testing, immersive interaction testing, and quiz feature testing.

After testing is completed, several improvements and bug fixes are carried out to improve application stability, feature performance, and overall user experience. This process is important to ensure that the application can provide effective and interactive educational experiences for users.

IV. Experimental Results

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